

Zhao Mingze

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OBJECTIVE

I am actively seeking research assistant, research internship, PhD, and such opportunities to deepen my expertise in AI and their applications in real-world scenarios. With a strong foundation in machine learning and a growing passion for reinforcement learning, I am eager to contribute to cutting-edge research projects while developing my technical and analytical skills. My long-term goal is to pursue a PhD in computer science. Although I am currently a Taught Postgraduate (TPG) student with limited opportunities to engage in research and minimal publications to date, I have never ceased seeking collaborative projects and am deeply committed to continuous learning.

EDUCATION

- **The University of Hong Kong** Sep. 2025 - In Progress
Master of Science in Computer Science
◦ Graduate with **distinction**.
◦ GPA: 3.641/4.0, NLP(A), Quantum Computing(A-), ML & RL(A-), Data Mining(A), AIoT(A).
- **Tongji University** Sep. 2021 - Jun. 2025
Bachelor of Engineering in Software Engineering
◦ GPA: 85/100, Third-class Merit Scholarship, School-level, Dec. 2022.

RESEARCH EXPERIENCES

- **Unifying Temporal and Structural Credit Assignment in LLM MAS Prompt Optimization** [🌐] Sep. 2025 - Aug. 2026
Research Assistant
◦ Drew the main table and the methodology figure. Drafted Introduction, Related Works, Methodology, Experiments and Conclusion.
◦ Complete 90% experimental verification. Deployed vllm using docker running 7-9B LLMs.
- **BIRD-CRITIC NL2SQL Dataset Alignment Annotation** [🌐] Sep. 2025 - Nov. 2025
Research Assistant
◦ Improve data quality of NL2SQL benchmark BIRD. Rewritten 300+ problematic SQL queries.
◦ Checked over 1500 instances systematically, identified and classified 6 common categories of data quality issues.
- **AI-Powered Interview Preparation Platform** Jan. 2026 - Aug. 2026
Postgraduate Project
◦ Integrating Intelligent Agents, Online Judge, and Personalized Learning.
- **A Large Model for Reasoning and Decision-Making based on Process Reward Models** Nov. 2024 - May. 2025
Undergraduate Thesis
◦ Used LLaMA-Factory to perform SFT and RLHF on Qwen2.5-7B-Instruct.

INTERNSHIPS

- **Cyber Security Company Anxinsec** [🌐] Jul. 2026 - In Progress
AI Dev Intern
◦ Responsible for the development of AI agent for the Red Team automated penetration testing platform.
◦ Integrated harness engineering methods, improving efficiency by 10% -50%.
- **NEV Company NIO Inc.** [🌐] Jul. 2024 - Oct. 2024
Low-Code Development Intern
◦ Developed a supply chain inventory analysis system, improving computing efficiency by 120%.
◦ Developed a backend in Nodejs, a frontend in JavaScript, and a database in MySQL.
◦ Conducted forecasts to validate future inventory.
- **Cyber Security Company Qianxin** [🌐] Jun. 2024 - Jul. 2024
Network Security Engineer
◦ Developed data-washing Python scripts, achieving 100% accuracy on gambling dataset.
◦ Conducted analysis on hardware, identifying criminal evidence, and presenting findings in documentations.

SKILLS

- **Fields explored:** base models (Transformers), systems (training & inference), RL, ...
- **Awards:** Chinese Mathematical Contest in Modeling 3rd prize
- **Tech stack:** C/C++, Python, JavaScript, MySQL, docker, PyTorch, sklearn, numpy, matplotlib, gradio, vllm, triton
- **Hobbies:** reading, gaming, writing blogs
- **Languages:** Mandarin (Native), English (Fluent), French & German (Beginner), Cantonese (Beginner)
 - IELTS 7 : Reading 8.5, Writing 6, Listening 6.5, Speaking 6
 - GRE 314 : Verbal 147, Quantitative 166, Writing 3.5

PROJECTS

• Stanford Spring 2026 CS336: Language Modeling from Scratch

Dec. 2025 - In Progress

Skills: PyTorch, nsys, cuda, triton, RL

Repo[[🔗](#)] Blog[[🔗](#)]

- Trained a tokenizer using TinyStories and OpenWebText datasets. Trained a 0.017B Transformer LLM.
- Constructed Transformer with RMSNorm, RoPE, SwiGLU.
- Explored single GPU nsys, activation checkpointing, operator fusion, and triton.
- Explored scaling law, data filtering, deduplication, SFT, DPO, and GRPO.

• MIT 6.S184 Introduction to Flow Matching and Diffusion Models 2026

Dec. 2025 - In Progress

Skills: PyTorch, numpy, matplotlib, differential equations

Blog[[🔗](#)]

- An exploration of diffusion large language models.

• Performed experiments for papers during RA

Sep. 2025 - Aug. 2026

Skills: DSPy, vllm, docker

Paper[[🔗](#)]

- To solve the credit allocation problem in multi-agent collaboration with large models, where only the final result is available and it is difficult to locate who/which round went wrong.
- Designed two mechanisms to respectively identify the poorly performing rounds and agents: 1. Aggregation module recording the intermediate status of each round for time credit. 2. Fixed role prompts for structural credit.
- Used DSPy to optimize prompt, only modify weak parts, and do not modify all prompts.
- Deployed vllm using docker running 7-9B LLMs. Wrote a script that enabled optimal GPU usage.
- Collected results, applied data mining, ran ablation tests.
- Experiments have shown that this method converges faster and performs better than black box optimization, and can improve the accuracy of multi-agent inference. However, the disadvantage is that it relies on complete interaction trajectories and has high overhead for long sequences.

• Postgraduate project: AI-Powered Interview Preparation Platform

Jan. 2026 - In Progress

Skills: Java, python, Springboot, LangGraph, harness engineering

Blog[[🔗](#)]

- Purpose: 1. Analyze and analyze resumes 2. Generate personalized learning plans 3. Generate daily knowledge points and code questions 4. Simulate interview Q&A.
- Tech stack: 1. RAG - MinIO + Qdrant + ElasticSearch (RRF) 2. Agent - LLM + prompt + tools + skills + MCP + memory + RAG 3. Loop - ReAct, Plan Execution, DAG.
- System Design: Technical Document $\subseteq$ Technical Stack $\subseteq$ Position $\subseteq$ Job Category. Client with an OJ.

• A demo AI agent CLI

Jun. 2026 - Jul. 2026

Skills: python, LangGraph, harness engineering

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- Architecture: 1. Sequential - Serial Router 2. Fanout - Parallel Read Only 3. Tree - Three Layers of Different Permissions 4. DAG - LLM Task Decomposition + Topology Sorting + Dynamic Appending.
- LangGraph: 1. Checkpoint - graph snapshot + execution location + timestamp 2. HITL - automatic or approval gating + key decision point 3. complexity gating 4. hierarchical model routing.
- Highlights: 1. Two level compression + anti-shaking 2. Docker sandbox + downgrade 3. RepoMap 4. SQLite memory system 5. Workplace isolation 6. Permission management + retry.

• COMP7604 Game final project: A 2D platformer rouge-like game

Jan. 2026 - May. 2026

Skills: C#, Unity, claude code

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- A trial of pure vibe-coding big project, with 1B tokens consumption.

• COMP7310 AIoT final project: A step counter based on mobile IMU sensors

Jan. 2026 - May. 2026

Skills: PyTorch, numpy, scipy, mobile sensors

Repo[[🔗](#)]

- Implemented offline batch processing, real-time online step detection, configurable signal processing algorithm (LPF, Hampel, Savitzky-Golay, CMA, ACF, FSM), and optional LSTM deep learning model.

• COMP7607A NLP final project: Musicode

Sep. 2025 - Nov. 2025

Skills: PyQT, Vue3, harness engineering

Blog[[🔗](#)]

- Team leader, responsible for the idea. Built up a simple role-based multi-agent system for generating MIDI.
- Other team members implemented intention recognition, frontend, and OCR.

• COMP7404 ML & RL mid-term project: Junk mail classification system

Oct. 2025 - Dec. 2025

Skills: Scikit-learn, nltk, JupiterNotebook, machine learning

Blog[[🔗](#)]

- Collected junk mail data, preprocessed it, performed feature engineering, trained models, and evaluated models.
- Models trained: Perceptron, Logistic Regression, KNN, SVM, Decision Tree, Random Forest.
- Metrics evaluated: precision, recall, f-measure, confusion matrix.

• COMP7404 ML & RL final project: Implementation of a paper and talking on a conference

Oct. 2025 - Dec. 2025

Skills: PyTorch, Scikit-learn, Transformers, machine learning, talking ability

Repo[[🔗](#)]

- Original paper: BiScope: AI-generated Text Detection by Checking Memorization of Preceding Tokens. [[Arxiv](#)]
- Modified LLM call, added local LLM inference. Developed a frontend based on their open-source Python scripts.